Remaining schematic work

PetSafe 100-1339 | v0.9.46 | 2026-10-15

10 active issues. Completed E11 results: C20 lower to U3 pin 3 = 1.023 Mohm, to TP7 = 998 kohm, and to R28 right = 1.5 ohm. The confirmed D2/C20/R28 group is now separate from U3.3. No new readings requested for this correction. Q3 stays fitted. Completed steps are not new bench requests.

Drawing / connectivity	Remaining data	ERC
0 fitted open pads; 0 open PIC pads; 0 dangling wire ends. 83 native/model net partitions match.	44 ceramic values; LED1 footprint blank. TP11/TP17 hidden-layer limit retained; unsupported R33 withdrawn.	1 retained U6/U6A output conflict. Six documented power-source flags; R5 supplies U2/C7 from RF rail.

Audit boundary

Native wire geometry, library definitions, all catalog pad assignments and existing evidence were checked. Zero open pads does not establish hidden copper or functional behavior. U6A is DNP; its regulator symbol and visible conflict remain by user choice. B59 supports the RA0-TP6 net merge. No new NC markers or ERC exclusions.

Order of work

Audit connection evidence and device qualifications: R39 and receiver-input split are corrected. E11 retains specific upper-capacitor/input and other copper-evidence gaps; no new bench batch. Review TOPOLOGY_AUDIT.md and E11 before further hardware work; retain deferred scope/isolation choices.

Recover critical component values: All 44 unknown ceramics have numerical design candidates, ratings and dielectrics: 12 supply, 17 signal/timing and 15 RF-bank seeds. Validation remains; no broad measurement batch is active. Review the function-based candidate table. No new bench batch now; prioritize shared filter parameters and antenna behavior over 44 individual readings.

Finish interface and package mapping: LED colour branches and J5 board-contact/output mapping are established. LED footprint, connector fit and external motor harness orientation remain reproduction qualifications. Obtain a matching LED vendor drawing or pad geometry; qualify J5 and the other packages before laying out a replacement PCB.

Audit connection evidence and device qualifications

E03 - RF supply and Q8 drive

Established: C5/C6 and Q8 source/ground/outer-drain joins supported. U2/C7 supply now fed from H_RF_VDD through R5 (22 ohm): local and endpoint readings B78-B81 support this route. U2 supply = 6.3 V; direct difference to printed VDD = +1.8 V rejects old VDD tie. Q8 output-to-ANT1 accepted from visible copper and compatible B82 4.5-ohm reading with antenna fitted. Rest gate-source voltages: P=0 V, N=+6.3 V. TI lists C14R among the DBV candidate markings; absolute maximum VCC is 6.5 V, distinct from the 5.5-V recommended maximum.

Remaining: Exact U2 identity and compatibility with measured 6.3 V remain unresolved; candidate SN74LVC2G14 recommended supply maximum is 5.5 V. Q8 switching margin, timing and RF waveform validation are deferred by owner. B82 alone is not a zero-ohm proof; output route is photo-supported.

Next: Photo and device-consistency review completed in TOPOLOGY_AUDIT.md. Keep the measured 6.3 V U2 feed and explicit 5.5 V recommended-limit conflict; no justified device replacement or further copper correction. None now. Scope and component-isolation work are deferred by owner.

Close when: U2 identity and voltage compatibility resolved; RF operating waveform validation recorded separately when the owner resumes scope work. Local RF supply/output drawing is corrected; unknown C6/C7 values remain V01.

E04 - Antenna and tuning-cell topology

Established: PIC-to-control resistors and local tuning midpoints are mapped. B44/B45 midpoint-to-antenna DC readings are OL. B52 C11 outer end to ANT2 = 1.4 ohm; photo-shared outer pads of C14/C29/C44/C47 now join ANT2. Rear capacitor returns remain GND. ANT1-ANT2 is about 5 ohm with antenna fitted. The mistaken 414 kohm N-B report was corrected to B-G. Q3 L-S latest 0.48 V, L-R 2.3 V; all other diode directions OL. Antenna inductance now reported as 500 uH; test conditions not specified. Existing V090/095/100/103/107 owner reports reconciled individually with original rear lands in IMG_2442 now support all five front/rear midpoint links. Seven OL midpoint exclusions remain.

Remaining: Tuning transistor family, native pin-role interpretation and switching behavior remain unresolved. The five midpoint correspondences now have owner-report plus original-photo support; this does not validate the PNP device model or capacitor estimates.

Next: Keep copper and starred values unchanged. Preserve Q3 diode results and qualify the device model independently of the now-supported midpoint joins. Scope and component isolation remain deferred; no owner test requested.

Close when: Tuning device pin roles and switched RF behavior are supported; outer ANT2 path is already resolved.

E07 - PIR: Q2 behavior and operating supply

Established: J3/U7 ordering, U6 external pads and PIC21/TP16/Q2 output routing are established. U6 pin5 is tied to VIN; pin4 is unused. B58 measured 3.3 V at J3.1 with PIR connected; logic VDD was 4.5 V. R39.1 joins TP16/Q2.S/PIC21 by continuous copper in original IMG_2425; R39.2 remains VDD. Earlier base-side assignment is withdrawn.

Remaining: Q2 terminal functions, PIR raw-signal behavior and PIC input response remain unverified. The loaded PIR supply is established at 3.3 V in the reported state; transients and other states are not characterized.

Next: Photo and device-consistency review completed in TOPOLOGY_AUDIT.md. R39 endpoint correction and native regression completed; Q2 identity and enabled behavior remain qualified. None now. Scope and component-isolation work are deferred by owner.

Close when: Q2 signal behavior and U6 output are supported by device or operating evidence.

E08 - Receiver and Q7 operating behavior

Established: U3 input returns and R23/R42/C40, C23/R25/R48 and DNP R47/C42 paths are reconciled. VREF supplies U3 pin5; C25.1-TP6.1 measured 1.5 ohm joins the receiver filter to PIC2/RA0. Powered: VREF 0 V, TP5 0.066 V, R19 left/right 0/4.5 V. Both R19-to-Q7 joins measured 1.5 ohm. Q7 diode results: R->L 0.74 V, L->R 1.74 V, S->L 0.74 V, L->S OL. These support PNP polarity with L as base; reverse finite reading can include parallel-board paths. B71 direct QR-QL measured 0 V, consistent with no forward base-emitter bias in the latest state and the low receiver supply; stability not reported.

Remaining: Why the receiver is inactive and its enabled-state behavior remain unverified. Earlier separate 0/4.5 V readings conflict with the later direct 0 V; do not diagnose failure or assume simultaneous identical states. Exact Q7 identity and E/C selection remain qualified. Receiver gain/filter response and MCU interface roles are unverified; DC readings do not exclude pulses.

Next: Photo and device-consistency review completed in TOPOLOGY_AUDIT.md. Feedback paths corroborated; inherited detector-to-U3.3 and some supply associations are now separately tracked in E11. Do not combine nonsimultaneous powered readings. None now. Scope and component-isolation work are deferred by owner.

Close when: Receiver supply, bias and the existing RA0/RA1/RB4 interfaces have supported operating roles; no presumed extra output wire is required.

E09 - Supply voltages and Q1 control

Established: VSYS/RF 6.3 V, printed VDD 4.5 V, PIR supply 3.3 V with PIR attached; VREF 0 V and red LED approximately 1 Hz. TP14 is on U5.8/OUTA at 1.5 ohm, with R43.2/V015; BATTERY+ stays separate. Receiver TP5 = 0.066 V (B62). R44 = 15 kohm (B83); TP14-to-Q1 presumed gate recheck = 21.4/21.7 kohm (B84/B85). The earlier 516-ohm conflict is not reproduced and no wiring change is warranted.

Remaining: Q1 fitted identity, terminal roles and powered control behavior remain unverified. The resistance recheck is compatible with R44 in the path but does not prove an intact MOSFET or the exact remaining path. Q1/motor dynamics and R3/R4/C28 sensing role remain qualified.

Next: Photo and device-consistency review completed in TOPOLOGY_AUDIT.md. R44 local route corroborated; 516-ohm contradiction stays closed by B84/B85. Q1 roles/control remain qualified. None now. Scope and component-isolation work are deferred by owner.

Close when: Actual supply levels and control behavior are recorded without merging battery, VSYS or filtered branches.

E10 - D6 breakdown and ICSP compatibility

Established: D6 anode is GND and cathode is VPP; forward drop is about 0.7V and reverse diode mode is OL. J1 pin order is resolved. Primary programming specification DS40001753B requires 8.0-9.0 V for high-voltage entry; low-voltage entry exists only when LVP is enabled.

Remaining: G3 is nonunique: MMSZ5228BS is 3.9 V; PZU2.4B is 2.4 V. Neither rating has been measured on this board. A directly connected clamp of either rating would load the required high-voltage ICSP level. The current PIC LVP configuration and actual D6 breakdown are unknown.

Next: Photo and device-consistency review completed in TOPOLOGY_AUDIT.md. Measured polarity is distinct from unmeasured breakdown; no safe in-circuit identification step is justified here. None now. Scope and component-isolation work are deferred by owner.

Close when: Clamp class/rating is established or explicitly bounded for reconstruction, and programming compatibility is checked.

I01 - Candidate device identities and pin roles

Established: Manufacturer candidate datasheets and pin tables exist. D2 local wiring is complete; measured R -> S -> L diode behavior supports a series pair.

Remaining: Exact fitted identity/ratings remain candidates for D1-D6, Q1-Q8/Q11, U1/U2/U6. The 4P marking alone does not identify three-pin D2; a manufacturer 4P code for a two-pin diode is incompatible.

Next: Photo and device-consistency review completed in TOPOLOGY_AUDIT.md. D2 series-junction equivalent documented separately from its unchanged physical placeholder. Primary U2/Q3 sources checked; remaining identities stay candidates. None now. Scope and component-isolation work are deferred by owner.

Close when: Each part has a supported pin/function model and adequate ratings or an explicit accepted-equivalent limitation; exact branding is not falsely claimed.

E11 - Inherited copper joins without independent closure

Established: C20/D2/R28 split is supported by the completed 1.023 Mohm, 998 kohm and 1.5 ohm results. Existing originals and via reports now close LED-control, crystal, PIC24/R4, L2-to-C1, R44/TP4, VSYS/C27/R3, C2-ground and five rear-midpoint coverage gaps. C17/R21/C24/U3.8 receiver supply subgroup is visible. No values or other wiring changed. Selected C27, PIC VSS/C32, Q9/R32/C41 and four tuning-pad ground branches also reconciled with existing owner via reports.

Remaining: Still unproved: C19 lower/model1-U3.3; C20 upper/model1-GND; U1 lower-left/native3-VIN; R18 upper/model2-Q7.L; R20.2/TP5 to C24.1 receiver supply; R3.2/R4.1/C28.1 junction; U5.1 and J1.VDD to printed VDD. Remaining ground branches are individually enumerated in TOPOLOGY_AUDIT.md. Missing selected evidence is not a demonstrated wiring fault.

Next: Existing-evidence review completed for this round; keep each unresolved physical link qualified. TOPOLOGY_AUDIT.md gives exact endpoints, original images and possible discriminating checks. Device/operating questions stay in their own issues. None requested in this round. C20 lower-to-U3.3 and its TP7/R28 follow-ups are completed. Do not repeat them; no automatic follow-up campaign, lifting or scope request.

Close when: Every remaining physical link has direct photo/owner evidence or a supported correction. The five midpoint evidence gaps are closed, but E11 as a whole remains open; documentation or deferred tests do not close it.

Recover critical component values

V01 - Capacitor values and passive ratings

Established: C5 is 470uF/16V; L1/L2 reported 1.4/2.2uH. Marked resistors are recorded. All 44 unknown ceramics now have numerical design candidates: 12 supply, 17 signal/timing and 15 tuning-bank parts, with proposed ratings/dielectrics. PetSafe variant FAQ supports 134.2 kHz. C21/C22 raw auto-mode readings remain preserved. Native v0.9.44 now displays all 44 estimates with asterisks. The owner reports antenna inductance of 500 uH; test conditions are unspecified.

Remaining: The 44 original fitted ceramic values remain unconfirmed. Numerical choices are explicit design proposals. The RF bank uses the reported 500 uH and assumes ideal switches. Inductance test conditions, weights/order, voltage stress and firmware tuning remain unverified. Receiver/timing choices and L1/L2 ratings also need qualification.

Next: Preserve all 44 starred estimates and their qualifications. Capacitor selection is outside this topology task. None for this task; value/package qualification remains later work.

Close when: Candidate choices are validated for the intended reconstruction; original-value and original-rating claims remain separate unless independently established.

All unrecovered ceramic values

C1, C2, C6, C7, C8, C9, C10, C11, C12, C13, C14, C15, C16, C17, C18, C19, C20, C21, C22, C23, C24, C25, C26, C27, C28, C29, C30, C31, C32, C33, C34, C35, C36, C37, C38, C39, C40, C41, C43, C44, C45, C46, C47, C48

Finish interface and package mapping

M01 - LED footprint and mechanical qualification

Established: LED1 is the only unassigned footprint; R30/PIC23 lights red and R31/PIC22 lights green. C5 diameter/height, S1 body/actuator size, R8 and C11 sizes are already measured. J5 rear placement and both board contacts are mapped to U5 outputs from front/rear photos.

Remaining: LED1 land pattern/manufacturer numbering and internal pairing of its externally grounded cathode pads remain unverified. C5 pitch and S1 pad fit/3D heights are reproduction qualifications, not unknown schematic wires. J5 stock-footprint pad order/pitch and external motor harness colours remain unverified.

Next: Preserve measured body dimensions and candidate footprints. LED1 pad geometry and exact fit remain reproduction prerequisites, outside this task. None for this task; value/package qualification remains later work.

Close when: LED footprint has a supported pad map; remaining package-fit assumptions are documented separately from schematic connectivity.

Documented reconstruction limitations

E06 - PIC14/TP11 and PIC18/TP17: hidden continuation not excluded

Disposition: Drawing resolved; physical completeness unproved

PIC14/RC3 reaches TP11; PIC18/RC7 reaches TP17. Front/rear photos show no onward surface route. Both local routes are represented and are valid endpoints.

Unproved: Additional inner-layer branches and firmware purpose remain unverified; test-only use is not asserted.

Next: None justified for E06. Do not repeat the rejected TP16-TP11 or TP16-TP17 checks.

Reopen if: A specific unassigned destination, contradictory continuity result, inner-layer image or firmware evidence identifies an onward connection that the current drawing omits.

evidence/e06_disposition_v0941.json

PCB reproduction requirements

Electrical design

Resolve the active main-board issue groups, recover critical values, and qualify candidate pin functions/ratings. TP11/TP17 hidden-layer completeness remains a documented reconstruction limitation.

Agent: Trace visible copper, review datasheets and keep the model/native files consistent.

Owner input: Antenna outer-rail continuity complete. Future measurements are selected individually after photo review; powered behavior remains unverified.

Mechanical fit

Dimensioned board perimeter, mounting holes, connector positions and lead geometry; LED land pattern.

Agent: Create a dimensioned replacement layout from supported measurements.

Owner input: Only geometry not recoverable reliably from the photos; existing body dimensions are retained.

Copper and stackup

No routed PCB exists; four layers remain a hypothesis. Continuity identifies nets, not hidden track geometry or dielectric thickness.

Agent: Design and check a replacement routing/stackup once electrical and mechanical data are sufficient.

Owner input: Layer/board thickness evidence if exact physical replication is required.

Firmware and configuration

No PIC firmware/configuration backup is present. An identical populated board needs the program and configuration, or compatible replacement firmware.

Agent: Plan a read-only backup after ICSP/D6 compatibility is established; do not erase the original.

Owner input: Programmer availability later. Protection may prevent reading the program; this is currently unknown.

Complete working assembly

Powered validation, motor/antenna/PIR integration, and daughterboard internals if that board is also being copied.

Agent: Define staged checks against the original and validate the replacement.

Owner input: Operating measurements on the original; external assemblies and their geometry if reproduced.

Scope and sources

Main-board schematic finalization. Routed PCB/layer stack, PIR daughterboard internals and firmware reconstruction remain later work, not completed. Empty-option population design is not needed to reproduce the observed DNP board.

DNP options retain all modeled local routes and candidate roles. J6.2/R36.1 terminates at an unpopulated external interface; no additional onboard rail is assumed. U6A stays a regulator symbol with its visible output conflict by explicit user choice.

All previous issue dispositions: evidence/finalization_review_v0931.json. Native geometry and uncertainty coverage: evidence/finalization_audit.json. Current register: evidence/remaining_work.json. Archived observations and old checklists: docs/history/README.md.

Power-source flags describe supply paths through existing passive components; they do not prove voltage or transistor operation. Installed KiCad 10 libraries and project symbols match the embedded pin/graphic definitions. GUI library reload has not been verified.

Reference rails

Rail	Existing modeled path
BATTERY+	Actual positive battery terminal -> Q1 single pad/native3 (candidate drain). TP14 is OUTA, separate from this rail.
VSYS	Q1.L/native2 -> U5/MX512H pin4; L1.1 and L2.1 supply sides.
VDD	TP103, PIC20/C32, U5 pin1/VCC, V064/V063/V075/V082. D3.R via V082 is user-confirmed.
VREF	TP104, PIC4/RA2/V035/C39, R21.2/V071. U3.5 and R22 D/left/model2 at 1 ohm.
GND	Battery negative and ground copper.